

Telecom Churn Data Analysis

Objective

The objective of this analysis is to explore customer churn behaviour in a telecom company by analysing demographic, usage, and subscription patterns. The goal is to:

- Understand the drivers behind customer churn (26.54%)
- Identify at-risk customer segments
- Suggest data-backed strategies to reduce churn and improve customer retention.

Statistics and Highlights

Metric	Value
Total Customers	~7,043
Churned Customers	~1,867 (26.54%)
Non-Churned Customers	~5,176 (73.46%)
Churned Senior Citizens	~42% of all senior citizens
Churned with Fiber Optic Internet	~49%
Churned with No OnlineSecurity	~35%
Churned with No TechSupport	~38%

Detailed Analysis by Feature

Demographics

- Senior Citizens (binary flag):
Senior citizens exhibit a significantly higher churn rate (~42%) compared to non-seniors.
- Partner/Dependents:
Customers with no partner or dependents tend to churn more — possibly due to lack of bundled/shared needs.

Service Features

1. Tenure (Months of Service)

- Customers with tenure < 12 months account for over 60% of total churn.
- In contrast, customers with tenure > 36 months have a churn rate of less than 10%.
- Interpretation: New users are more volatile and require stronger onboarding.

2. Internet Service

- Fiber Optic users churn the most: nearly 49% of fiber users churned.
- DSL users have moderate churn, while users with no internet service have the lowest churn (~8%).

3. Online Security / Backup / Device Protection / Tech Support

- Customers not subscribed to these services churn at a 30–38% rate.
- Those using these features show lower churn (10–15%).
- Conclusion: These value-added services are strong loyalty drivers.

4. Streaming Services (TV/Movies)

- Streaming features have a moderate effect:
 - Churn among non-streaming users: ~25–30%
 - Churn among streaming users: ~20–22%
- Streaming services alone aren't strong churn reducers, but combined with other features, they add value.

Contract and Billing Insights

- Month-to-month contracts have a churn rate of 43%, the highest among all.
- One-year and two-year contracts have significantly lower churn (~10–15%).
- Electronic Check payment method is associated with a much higher churn (38%), indicating possible financial friction.

What Can Be Done to Reduce Churn?

1. Improve First 3-Month Experience

- Focus on customers with tenure < 3 months by:
 - Proactive onboarding (emails, tutorials, concierge calls)
 - Early engagement offers
 - Surveys to catch early dissatisfaction

2. Upsell Retention-Friendly Services

- Promote OnlineSecurity, TechSupport, and DeviceProtection through:
 - Free trials in the first 2 months
 - Bundled pricing or “value packs”
 - In-app notifications about protection benefits

3. Improve Fiber Optic User Satisfaction

- Investigate customer complaints and technical issues with Fiber plans.
- Prioritize QoS (Quality of Service) initiatives for fiber users.
- Offer tech support/assistance to improve their experience.

4. Contract Incentives

- Move more users to 1- or 2-year contracts through:
 - Loyalty discounts
 - Early renewal perks
 - Cashback or reward systems for auto-renewal

5. Ease Payment Friction

- Reduce churn from electronic check users:
 - Educate on auto-debit, digital wallets
 - Offer small discounts for switching to auto-pay

Conclusion

This analysis reveals that customer churn is not random — it is highly influenced by tenure, contract type, service usage, and support access. By focusing on early retention, feature bundling, and service quality, the company can reduce churn and increase lifetime value.